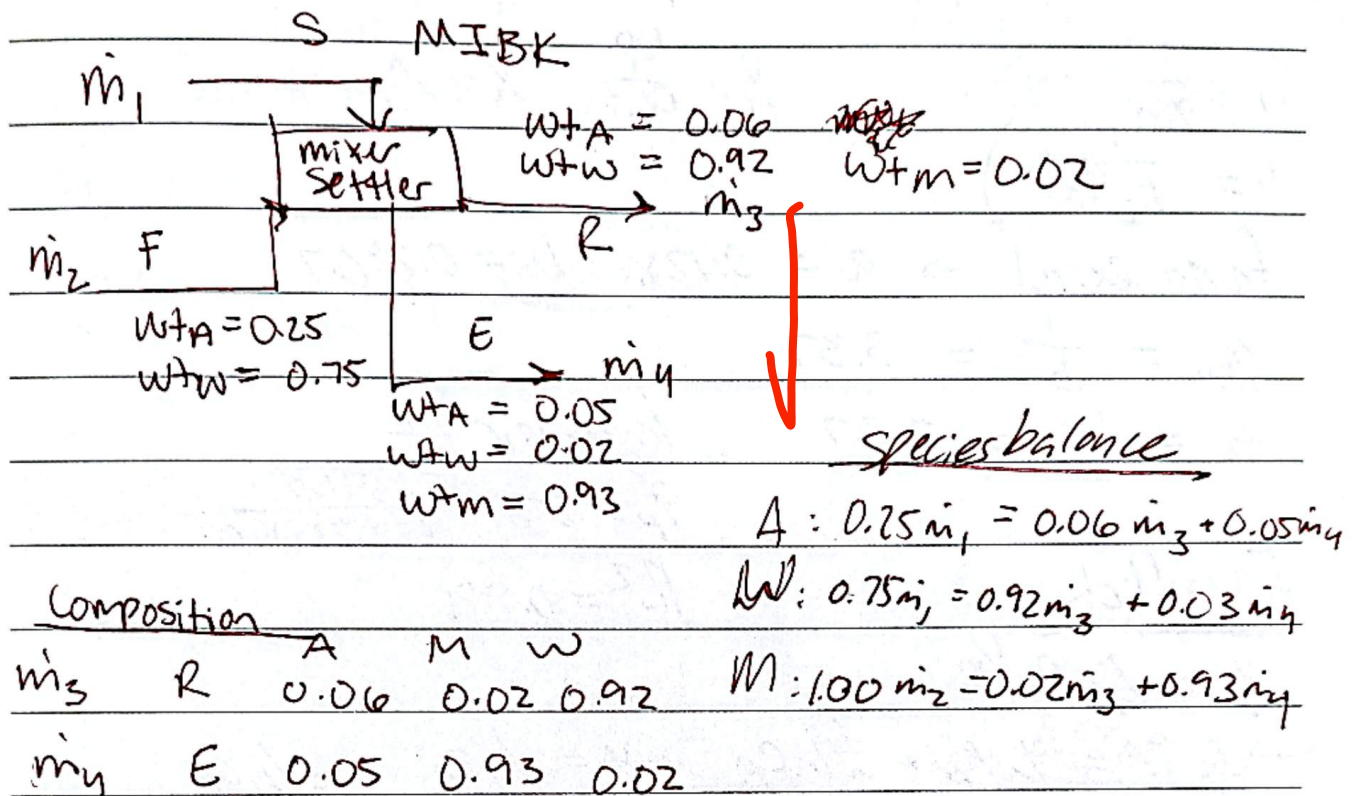


CHE 210 HW #9

Problem 1: Murphy 5.40



Basis: $100 \text{ kg/hr} = \dot{m}_1$

$$\begin{bmatrix} 0.06 & 0.05 \\ 0.92 & 0.03 \end{bmatrix} \begin{bmatrix} \dot{m}_3 \\ \dot{m}_4 \end{bmatrix} = \begin{bmatrix} 25 \\ 75 \end{bmatrix}$$

$\dot{m}_3 = 67.87 \text{ kg/hr}$

$\dot{m}_4 = 418.55 \text{ kg/hr}$

$$\dot{m}_2 = (67.87 \times 0.02) + (418.55 \times 0.93) = 390.61 \text{ kg/hr}$$

$$\frac{\dot{m}_2}{\dot{m}_1} = \frac{390.61}{100} = 3.91 \frac{\text{kg solvent}}{\text{kg feed}}$$

29125

Problem 2 : Murphy 5.91

$$q_p = q_{\infty} \left(\frac{K_L C_p}{1 + K_L C_p} \right) \rightarrow \frac{C_p}{q_p} = \frac{1}{K_L q_{\infty}} + \frac{C_p}{q_{\infty}}$$

$$b = \frac{1}{q_{\infty}}, \quad y = \frac{C_p}{q_p}, \quad x = C_p$$

$$a = \frac{1}{K_L q_{\infty}}$$

from excel $\rightarrow a = 0.128 \quad b = 0.2967$

$$q_{\infty} = \frac{1}{b} = 3.37$$

$$K_L = \frac{1}{a q_{\infty}} = 2.37$$

Langmuir fit

$$q_p = 3.37 \frac{2.37 C_p}{1 + 2.37 C_p}$$

Frendlich

$$R^2 = 0.89$$

$$q_p = K_F C_p^{1/n}$$

$$\rightarrow \ln q_p = \ln K_F + \frac{1}{n} \ln C_p \quad y = \ln q_p \quad x = \ln C_p$$

$$\rightarrow A = \ln K_F \quad B = \frac{1}{n}$$

$$\text{slope: } B = 0.245$$

Frendlich Fit $\int$ int. : $A = 0.758$

$$q_p = 2.14 C_p^{0.245}$$

$$R^2 = 0.991$$

← Better data fit

$$q_p = K_F C_p^{1/n} = 2.14 C_p^{0.245}$$

$$V = 1.0 \text{ L}$$

$$m_s = 5.0 \text{ g}$$

$$MW = 94.11 \text{ g/mol}$$

$$n_{\text{total}} = \frac{1}{94.11} \times 100 = 10.63 \text{ mmol}$$

$$n_{\text{total}} = n_{\text{sol}} + n_{\text{ads}}$$

$$10.63 = V C_p + m_s q_p = 1 \times C_p + 5(2.14 C_p^{0.245})$$

$$10.63 = C_p + 10.7 C_p^{1/4.10}$$

Using $C_p = 0.73 \text{ mmol/L}$

amount absorbed

$$q_p = 2.14 (C_p)^{1/4.10}$$

$$n_{ads} = M_s q_p$$

$$= (2.14)(0.93) = 1.98$$

$$= (5)(1.98)$$

$$= 9.9$$

Phenol fraction

$$\frac{n_{ads}}{n_{tot}} = \frac{9.9}{10.63} = 0.93$$

25/25

~~Problem 3~~ Problem 3

$$25^\circ\text{C} \rightarrow 150^\circ\text{C}$$

$$\dot{V}_{air} = 1.25 \text{ m}^3/\text{min}$$

$$\Delta h = 3640 \text{ J/mol}$$

$$P = 122 \text{ kPa}$$

a) $\dot{W} = 0$ $\dot{Q} = \dot{n} \Delta h \rightarrow \dot{n} = \frac{P \dot{V}}{RT}$ $\dot{V} = \frac{1.25}{60} = 0.0208 \text{ m}^3/\text{s}$

$$\Delta KE = 0 \quad P = 122 \text{ kPa} = 122,000 \text{ Pa}$$

$$\Delta PE = 0 \quad T = 150^\circ\text{C} + 273 = 423 \text{ K}$$

$$\dot{n} = \frac{(122,000 \text{ Pa})(0.0208)}{(8.314)(423)} = 0.723 \text{ mol/s}$$

$$\dot{Q} = \dot{n} \Delta h = (0.723)(3640) = 2632 \text{ J/s} = 2.63 \text{ kW}$$

b) need more info on the outlet velocity, specifically if it is larger or smaller than the inlet velocity and ~~state~~ what the values for the cross-sectional areas are.

25/25

Problem 4: Murphy 6.26

$$\begin{aligned} \text{(a)} \quad \Delta E_k &= \frac{1}{2} m (V_f^2 - V_i^2) = \frac{1}{2} \left(\cancel{1 \text{ mol H}_2\text{O}} \times \frac{18.01 \text{ g H}_2\text{O}}{\cancel{1 \text{ mol H}_2\text{O}}} \right) (100^2 - 0^2) \left(\frac{1 \text{ m}}{10^{-3} \text{ m}} \right)^2 \left(\frac{1 \text{ h}}{3600 \text{ s}} \right)^2 \\ &= \frac{1}{2} \left(\frac{1 \text{ kg}}{10^{-3} \text{ kg}} \right) = 6.948 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = 6.948 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \Delta E_p &= m g (z_f - z_i) = (18.01 \text{ g H}_2\text{O}) (9.81 \text{ m/s}^2) (100 - 0) \\ &= 17.668 \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2} = 17.668 \text{ J} \end{aligned}$$

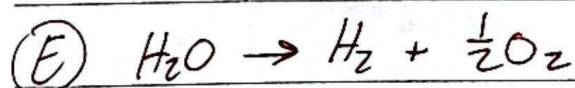
$$\begin{aligned} \text{(c)} \quad \Delta U &= m \Delta \hat{U} = (18.01 \text{ g H}_2\text{O}) (2506.2 - 2515.5) \text{ J/g} \\ &= -167.493 \text{ J} \end{aligned}$$

$$\hat{U}(100^\circ\text{C}, 0.1 \text{ barr}) = 2506.2 \text{ J/g}$$

$$\hat{U}(100^\circ\text{C}, 1 \text{ barr}) = 2515.5 \text{ J/g}$$

$$\begin{aligned} \text{(d)} \quad \Delta U &= n C_v \Delta T = (1 \text{ mol H}_2\text{O}) (75.4 \text{ J/g}) \\ &= (100 - 0)^\circ\text{C} = 7540 \text{ J} \end{aligned}$$

$$C_v \approx C_p \quad \text{from pg}$$



$$\begin{aligned} \Delta \hat{H}_r &= (0)(0) + (0)(0) - (1)(-285.85 \text{ kJ/mol}) \\ &= 285.85 \text{ kJ/mol} \end{aligned}$$

$$\begin{aligned} \Delta \hat{U}_r &= \Delta \hat{H}_r - R T \Delta n_r = (285.85 \text{ kJ/mol}) \left(\frac{1 \text{ J}}{10^{-3} \text{ kJ}} \right) \\ &\quad - (8.314 \text{ J/mol} \cdot \text{K}) (298 \text{ K}) \left(\frac{1}{2} \text{ mol} \right) \\ &= \Delta \hat{U}_r = 287,611 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{(f)} \quad \Delta E &= (18.01 \text{ g H}_2\text{O}) (2.9979 \times 10^8 \text{ m/s})^2 \left(\frac{1 \text{ kg}}{10^{-3} \text{ g}} \right) \\ &= 1.619 \times 10^{15} \text{ J} \end{aligned}$$